Monitoring storage capacity

StorageGRID 11.5

NetApp
February 20, 2024

This PDF was generated from https://docs.netapp.com/us-en/storagegrid-115/monitor/monitoring-storage-capacity-for-entire-grid.html on February 20, 2024. Always check docs.netapp.com for the latest.

Table of Contents

Monitoring storage capacity 1
Monitoring storage capacity for the entire grid 1
Monitoring storage capacity for each Storage Node 4
Monitoring object metadata capacity for each Storage Node 7

Monitoring storage capacity

You must monitor the total usable space available on Storage Nodes to ensure that the StorageGRID system does not run out of storage space for objects or for object metadata.

StorageGRID stores object data and object metadata separately, and reserves a specific amount of space for a distributed Cassandra database that contains object metadata. Monitor the total amount of space consumed for objects and for object metadata, as well as trends in the amount of space consumed for each. This will enable you to plan ahead for the addition of nodes and avoid any service outages.

You can view storage capacity information for the entire grid, for each site, and for each Storage Node in your StorageGRID system.

Related information

Viewing the Storage tab

Monitoring storage capacity for the entire grid

You must monitor the overall storage capacity for your grid to ensure that adequate free space remains for object data and object metadata. Understanding how storage capacity changes over time can help you plan to add Storage Nodes or storage volumes before the grid's usable storage capacity is consumed.

What you'll need

You must be signed in to the Grid Manager using a supported browser.

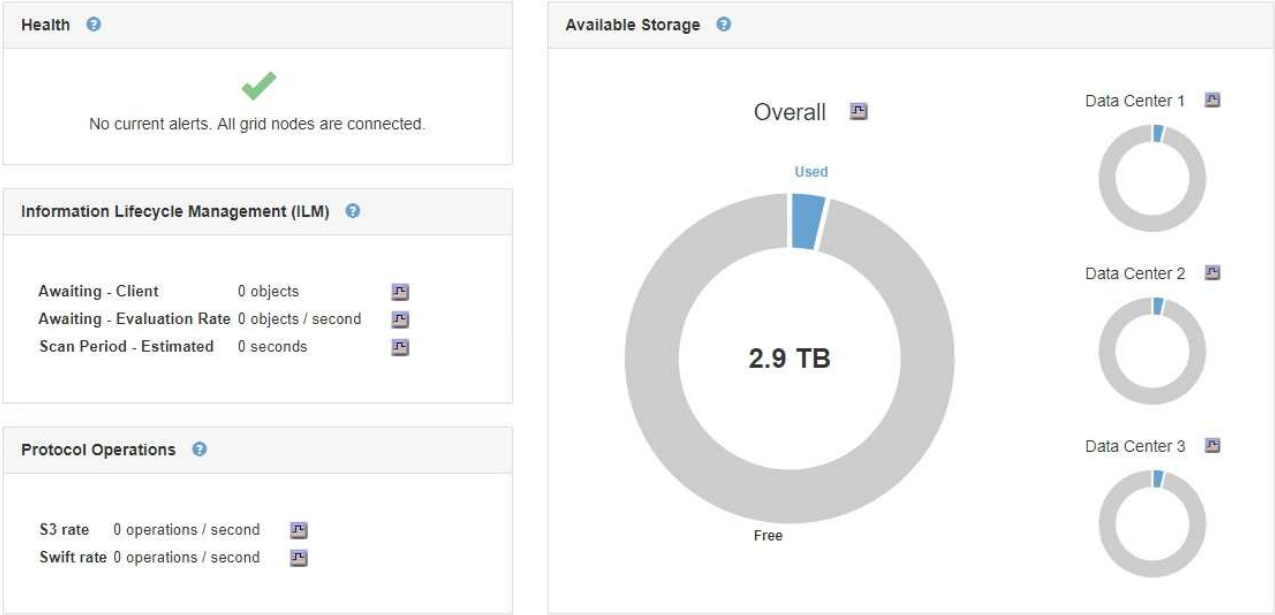
About this task

The Dashboard in the Grid Manager lets you quickly assess how much storage is available for the entire grid and for each data center. The Nodes page provides more detailed values for object data and object metadata.

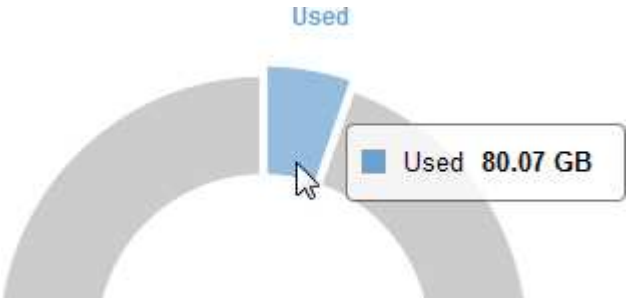
Steps

1. Assess how much storage is available for the entire grid and for each data center.
   a. Select **Dashboard**.
   b. In the Available Storage panel, note the overall summary of free and used storage capacity.

      The summary does not include archival media.



c. Place your cursor over the chart's Free or Used capacity sections to see exactly how much space is free or used.



d. For multi-site grids, review the chart for each data center.

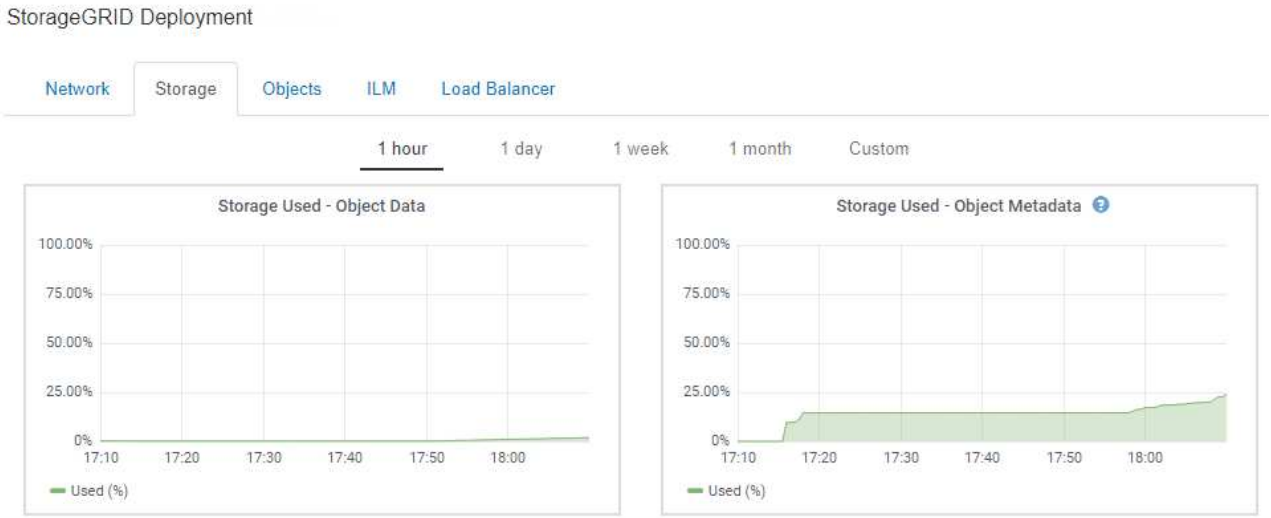
e. Click the chart icon for the overall chart or for an individual data center to view a graph showing capacity usage over time.

   A graph showing Percentage Storage Capacity Used (%) vs. Time appears.

2. Determine how much storage has been used and how much storage remains available for object data and object metadata.

   a. Select **Nodes**.

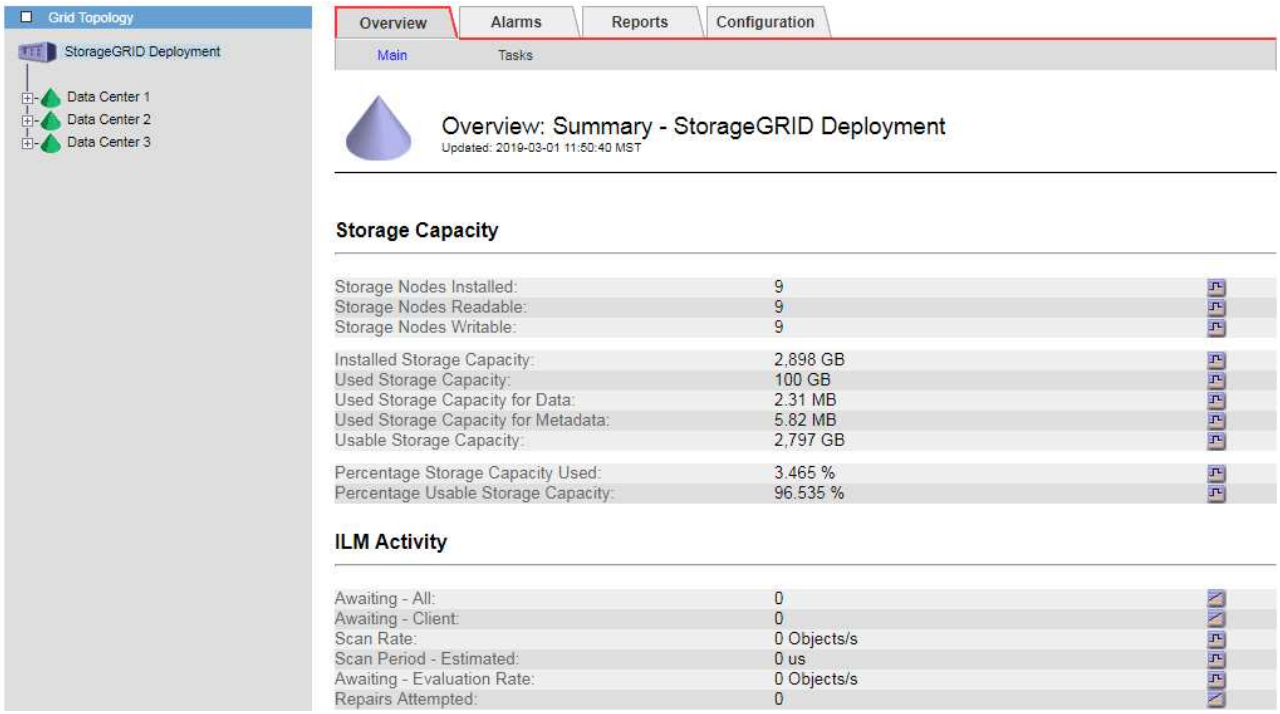
   b. Select ***grid*** > **Storage**.



c. Hover your cursor over the Storage Used - Object Data and the Storage Used - Object Metadata charts to see how much object storage and object metadata storage is available for the entire grid, and how much has been used over time.

> The total values for a site or the grid do not include nodes that not have reported metrics for at least five minutes, such as offline nodes.

3. As directed by technical support, view additional details about the storage capacity for your grid.
   a. Select **Support** > **Tools** > **Grid Topology**.
   b. Select ***grid*** > **Overview** > **Main**.



4. Plan to perform an expansion to add Storage Nodes or storage volumes before the grid's usable storage capacity is consumed.

   When planning the timing of an expansion, consider how long it will take to procure and install additional

storage.

> If your ILM policy uses erasure coding, you might prefer to expand when existing Storage Nodes are approximately 70% full to reduce the number of nodes that must be added.

For more information on planning a storage expansion, see the instructions for expanding StorageGRID.

Related information

Expand your grid

Monitoring storage capacity for each Storage Node

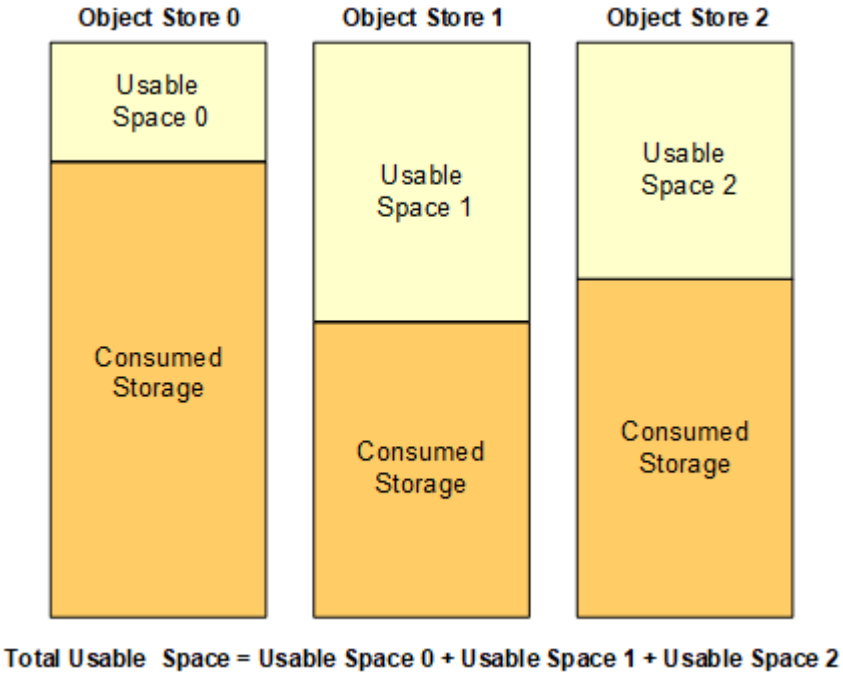
You must monitor the total usable space for each Storage Node to ensure that the node has enough space for new object data.

What you'll need

- You must be signed in to the Grid Manager using a supported browser.

About this task

Usable space is the amount of storage space available to store objects. The total usable space for a Storage Node is calculated by adding together the available space on all object stores within the node.



Steps

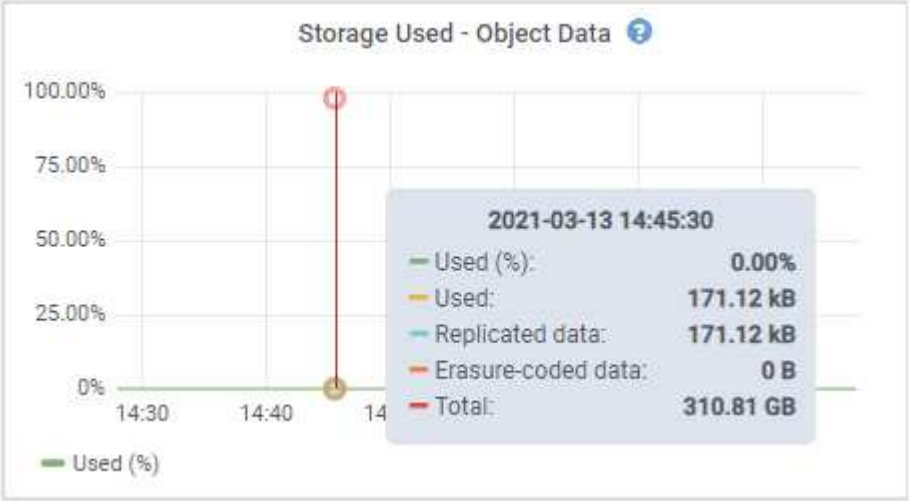
1. Select **Nodes** > ***Storage Node*** > **Storage**.

   The graphs and tables for the node appear.

2. Hover your cursor over the Storage Used - Object Data graph.

   The following values are shown:

- **Used (%)**: The percentage of the Total usable space that has been used for object data.
- **Used**: The amount of the Total usable space that has been used for object data.
- **Replicated data**: An estimate of the amount of replicated object data on this node, site, or grid.
- **Erasure-coded data**: An estimate of the amount of erasure-coded object data on this node, site, or grid.
- **Total**: The total amount of usable space on this node, site, or grid. The Used value is the `storagegrid_storage_utilization_data_bytes` metric.



3. Review the Available values in the Volumes and Object Stores tables, below the graphs.

To view graphs of these values, click the chart icons in the Available columns.

Disk Devices

Name	World Wide Name	I/O Load	Read Rate	Write Rate
croot(8:1,sda1)	N/A	0.03%	0 bytes/s	3 KB/s
cvloc(8:2,sda2)	N/A	0.85%	0 bytes/s	58 KB/s
sdc(8:16,sdb)	N/A	0.00%	0 bytes/s	81 bytes/s
sdd(8:32,sdc)	N/A	0.00%	0 bytes/s	82 bytes/s
sde(8:48,sdd)	N/A	0.00%	0 bytes/s	82 bytes/s

Volumes

Mount Point	Device	Status	Size	Available	Write Cache Status
/	croot	Online	21.00 GB	14.90 GB	Unknown
/var/local	cvloc	Online	85.86 GB	84.10 GB	Unknown
/var/local/rangedb/0	sdc	Online	107.32 GB	107.18 GB	Enabled
/var/local/rangedb/1	sdd	Online	107.32 GB	107.18 GB	Enabled
/var/local/rangedb/2	sde	Online	107.32 GB	107.18 GB	Enabled

Object Stores

ID	Size	Available	Replicated Data	EC Data	Object Data (%)	Health
0000	107.32 GB	96.45 GB	250.90 KB	0 bytes	0.00%	No Errors
0001	107.32 GB	107.18 GB	0 bytes	0 bytes	0.00%	No Errors
0002	107.32 GB	107.18 GB	0 bytes	0 bytes	0.00%	No Errors

4. Monitor the values over time to estimate the rate at which usable storage space is being consumed.
5. To maintain normal system operations, add Storage Nodes, add storage volumes, or archive object data before usable space is consumed.

   When planning the timing of an expansion, consider how long it will take to procure and install additional storage.

   > If your ILM policy uses erasure coding, you might prefer to expand when existing Storage Nodes are approximately 70% full to reduce the number of nodes that must be added.

   For more information on planning a storage expansion, see the instructions for expanding StorageGRID.

   The **Low object data storage** alert and the legacy Storage Status (SSTS) alarm are triggered when insufficient space remains for storing object data on a Storage Node.

Related information

Administer StorageGRID

Troubleshooting the Low object data storage alert

Expand your grid

Monitoring object metadata capacity for each Storage Node

You must monitor the metadata usage for each Storage Node to ensure that adequate space remains available for essential database operations. You must add new Storage Nodes at each site before object metadata exceeds 100% of the allowed metadata space.

What you'll need

- You must be signed in to the Grid Manager using a supported browser.

About this task

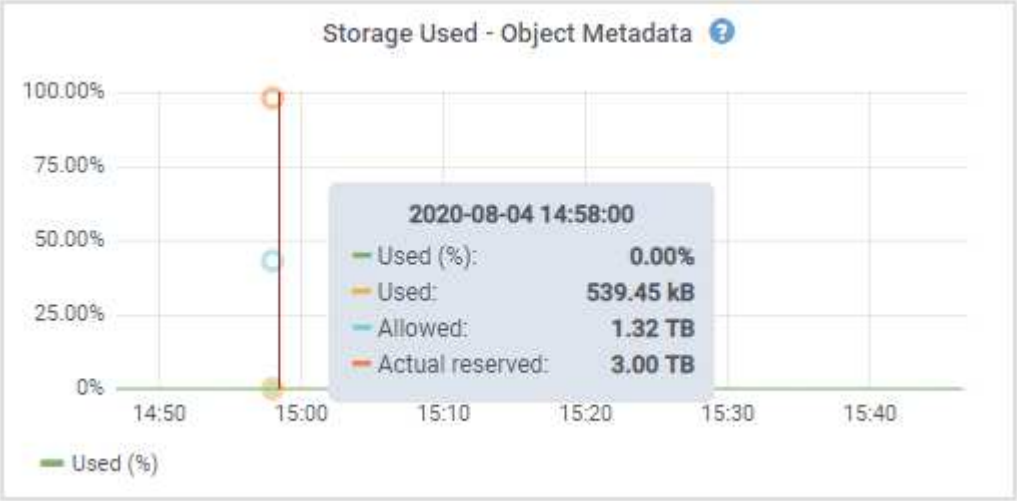
StorageGRID maintains three copies of object metadata at each site to provide redundancy and to protect object metadata from loss. The three copies are evenly distributed across all Storage Nodes at each site using the space reserved for metadata on storage volume 0 of each Storage Node.

In some cases, the grid's object metadata capacity might be consumed faster than its object storage capacity. For example, if you typically ingest large numbers of small objects, you might need to add Storage Nodes to increase metadata capacity even though sufficient object storage capacity remains.

Some of the factors that can increase metadata usage include the size and quantity of user metadata and tags, the total number of parts in a multipart upload, and the frequency of changes to ILM storage locations.

Steps

1. Select **Nodes** > ***Storage Node*** > **Storage**.
2. Hover your cursor over the Storage Used - Object Metadata graph to see the values for a specific time.



Value	Description	Prometheus metric
Used (%)	The percentage of the allowed metadata space that has been used on this Storage Node.	`storagegrid_storage_utilization_metadata_bytes/ storagegrid_storage_utilization_metadata_allowed_bytes`

Value	Description	Prometheus metric
Used	The bytes of the allowed metadata space that have been used on this Storage Node.	`storagegrid_storage_utilization_metadata_bytes`
Allowed	The space allowed for object metadata on this Storage Node. To learn how this value is determine for each Storage Node, see the instructions for administering StorageGRID.	`storagegrid_storage_utilization_metadata_allowed_bytes`
Actual reserved	The actual space reserved for metadata on this Storage Node. Includes the allowed space and the required space for essential metadata operations. To learn how this value is calculated for each Storage Node, see the instructions for administering StorageGRID.	`storagegrid_storage_utilization_metadata_reserved_bytes`

> The total values for a site or the grid do not include nodes that have not reported metrics for at least five minutes, such as offline nodes.

3. If the **Used (%)** value is 70% or higher, expand your StorageGRID system by adding Storage Nodes to each site.

> The **Low metadata storage** alert is triggered when the **Used (%)** value reaches certain thresholds. Undesirable results can occur if object metadata uses more than 100% of the allowed space.

When you add the new nodes, the system automatically rebalances object metadata across all Storage Nodes within the site. See the instructions for expanding a StorageGRID system.

Related information

Troubleshooting the Low metadata storage alert

Administer StorageGRID

Expand your grid

Copyright information

Copyright © 2024 NetApp, Inc. All Rights Reserved. Printed in the U.S. No part of this document covered by copyright may be reproduced in any form or by any means—graphic, electronic, or mechanical, including photocopying, recording, taping, or storage in an electronic retrieval system—without prior written permission of the copyright owner.

Software derived from copyrighted NetApp material is subject to the following license and disclaimer:

THIS SOFTWARE IS PROVIDED BY NETAPP “AS IS” AND WITHOUT ANY EXPRESS OR IMPLIED WARRANTIES, INCLUDING, BUT NOT LIMITED TO, THE IMPLIED WARRANTIES OF MERCHANTABILITY AND FITNESS FOR A PARTICULAR PURPOSE, WHICH ARE HEREBY DISCLAIMED. IN NO EVENT SHALL NETAPP BE LIABLE FOR ANY DIRECT, INDIRECT, INCIDENTAL, SPECIAL, EXEMPLARY, OR CONSEQUENTIAL DAMAGES (INCLUDING, BUT NOT LIMITED TO, PROCUREMENT OF SUBSTITUTE GOODS OR SERVICES; LOSS OF USE, DATA, OR PROFITS; OR BUSINESS INTERRUPTION) HOWEVER CAUSED AND ON ANY THEORY OF LIABILITY, WHETHER IN CONTRACT, STRICT LIABILITY, OR TORT (INCLUDING NEGLIGENCE OR OTHERWISE) ARISING IN ANY WAY OUT OF THE USE OF THIS SOFTWARE, EVEN IF ADVISED OF THE POSSIBILITY OF SUCH DAMAGE.

NetApp reserves the right to change any products described herein at any time, and without notice. NetApp assumes no responsibility or liability arising from the use of products described herein, except as expressly agreed to in writing by NetApp. The use or purchase of this product does not convey a license under any patent rights, trademark rights, or any other intellectual property rights of NetApp.

The product described in this manual may be protected by one or more U.S. patents, foreign patents, or pending applications.

LIMITED RIGHTS LEGEND: Use, duplication, or disclosure by the government is subject to restrictions as set forth in subparagraph (b)(3) of the Rights in Technical Data -Noncommercial Items at DFARS 252.227-7013 (FEB 2014) and FAR 52.227-19 (DEC 2007).

Data contained herein pertains to a commercial product and/or commercial service (as defined in FAR 2.101) and is proprietary to NetApp, Inc. All NetApp technical data and computer software provided under this Agreement is commercial in nature and developed solely at private expense. The U.S. Government has a non-exclusive, non-transferrable, nonsublicensable, worldwide, limited irrevocable license to use the Data only in connection with and in support of the U.S. Government contract under which the Data was delivered. Except as provided herein, the Data may not be used, disclosed, reproduced, modified, performed, or displayed without the prior written approval of NetApp, Inc. United States Government license rights for the Department of Defense are limited to those rights identified in DFARS clause 252.227-7015(b) (FEB 2014).

Trademark information

NETAPP, the NETAPP logo, and the marks listed at http://www.netapp.com/TM are trademarks of NetApp, Inc. Other company and product names may be trademarks of their respective owners.